

Island Convergence Report: MRX Pushforward(u_2) vs. SIMSOPT Biot–Savart

QUASR 0044970 · GVEC map · $n_{\text{fp}} = 3$ · Hodge $k = 2$ DBC nullspace

Primary script: `compare_push_u_to_simsomt.py`

27 July 2026

Purpose

This report has exactly two goals:

1. demonstrate convergence of the MRX vacuum field to the SIMSOPT Biot–Savart field; and
2. demonstrate convergence of the MRX island width to the physical SIMSOPT island.

Section 6 records the coding changes that were required to make those demonstrations reliable.

1 Problem setup

- **Equilibrium / coils:** QUASR serial 0044970 (`serial0044970.json`), volume mesh `quasr_new_0044970_mrx.h5`.
- **MRX field:** the Dirichlet $k = 2$ Hodge–Laplacian null form u_2 , pushed through the GVEC logical map as `Pushforward( $u_2$ )`.
- **Reference:** SIMSOPT Biot–Savart on the same coil set.
- **Resonance:** the physical $\iota = 1/2$, six-lobe chain near $\rho \approx 0.69$, with resonant content $(m, n) = (6, 3)$.
- **Comparison:** aligned relative vector L^2 , resonant normal-error amplitude, rotational transform, and Poincaré width.

2 Why the vacuum field converges

The MRX field is not fit to SIMSOPT data. It is the harmonic 2-form of the domain. In the finite-element exterior-calculus de Rham complex it lies in the $H(\text{div})$ space and is the Dirichlet nullspace of

$$\Delta_2 = d\delta + \delta d, \quad \Delta_2 u = 0. \quad (1)$$

For a 2-form, $du = 0$ and $\delta u = 0$ are respectively the solenoidal and current-free vacuum conditions. The nullspace dimension is the relevant Betti number and is one for this problem. FEEC refinement converges to the continuous harmonic form, while the Piola pushforward preserves the de Rham structure when mapping it to the physical magnetic field.

The discrete harmonic condition is auditable through

$$\langle u, \Delta_2 u \rangle = \|du\|^2 + \|\delta u\|^2. \quad (2)$$

The production ladder therefore records the Rayleigh quotient, relative divergence and weak-curl energies, the energy-identity residual, and the M_2 -metric comparison to the dense eigenvector.

Dense verification. The $k = 2$ Dirichlet nullspace is one-dimensional and separated from the rest of the spectrum by a large gap. Nevertheless, `find_nullspace_vectors` uses shift-and-invert inverse iteration whose inner inverse applications are inexact preconditioned-MINRES solves; harmonic coarse correction is disabled. The iteration can return a vector with high M_2 overlap with the nullspace but with an impure component that is not sufficiently harmonic for resonant topology. For every reported ladder point, dense `eigh` on the assembled Hodge Laplacian replaces that iterative result with the exact discrete harmonic form. The dense vector satisfies the discrete null equation and (2), restores monotone field convergence, and is the sole vector used for the reported diagnostics.

3 Metrics and definitions

Aligned relative L^2 . For equally weighted sample points p , the optimal global scale and minimum-error relative norm are

$$s^* = \frac{\sum_p u_p \cdot B_p}{\sum_p u_p \cdot u_p}, \quad L^2_{\text{aligned}} = \frac{(\frac{1}{N} \sum_p |s^* u_p - B_p|^2)^{1/2}}{(\frac{1}{N} \sum_p |B_p|^2)^{1/2}}. \quad (3)$$

Circulation normalization $\alpha = \Gamma(B_{\text{SIMSOPT}})/\Gamma(B_{\text{MRX,unit}})$ on the native one-field-period map agrees with s^* below 10^{-4} on resolved grids; a uniform nonzero scale leaves field-line trajectories unchanged, so we retain s^* because it defines the minimum-error L^2 .

Resonant normal-error amplitude. At the resonant surface ρ^* ,

$$a_{m,n} = \left| \left\langle (B_{\text{MRX}} - B_{\text{SIMSOPT}}) \cdot \hat{n} e^{-2\pi i(m\theta - (n/n_{\text{fp}})\zeta)} \right\rangle_{\theta,\zeta} \right|, \quad \hat{n} = \frac{\nabla \rho}{|\nabla \rho|}. \quad (4)$$

Here $(m, n) = (6, 3)$, so $a_{6,3}$ measures the error that directly drives the $\iota = 1/2$ chain and need not track global L^2 .

Rotational transform and resonance location. The unwrapped poloidal angle in cycles is fit against section-return index, giving $\iota = n_{\text{fp}} \times \text{slope}$. Interpolation of the $\iota = 1/2$ crossing gives ρ^* , around which the island seeds are centered.

Fourier-detrended physical width. For each traced line, $\rho(\theta)$ is fit to a constant plus poloidal modes 1–4. The $q_{95} - q_{05}$ spread after subtracting that smooth background removes surface-label shape. Convergence over SIMSOPT trace length, radial seeds, phase seeds, and tolerance gives $w_{\text{SIMSOPT,detrended}} = 0.0486\text{--}0.0503$, so $w_{\text{phys}} \approx 0.050$ is the quantitative physical-width reference. Re-inverting the dense SIMSOPT trace on the correct physical $\phi = 0$ section leaves that value unchanged at 0.0501.

Fixed-tracer trapped width. An orbit is classified as trapped when the six-lobe circular concentration $C = |N^{-1} \sum_i e^{2\pi i m \theta_i}|$ with $m = 6$ satisfies $C \geq 0.45$. Pooling trapped hits under one frozen radial $\times$ phase protocol gives $w_{\text{trapped}} = \rho_{q95} - \rho_{q05}$. This metric separates numerical over-islands from resolved grids under an identical tracer, but SIMSOPT itself returns 0.0197–0.0408 as radial and phase seeding change, so it is not claimed to be seed-converged or universal.

Table 1: Selected frozen-tracer FEM results ($16\rho \times 6\theta$; 2000 turns except $12 \times 24 \times 12$ at 1000). w_{det} is the maximum Fourier-detrended line width; w_{trap} is protocol-specific.

Grid	$n_{2,\text{dbc}}$	Aligned L^2	$a_{6,3}$	w_{det}	w_{trap}
$8 \times 16 \times 8$	2192	3.97×10^{-3}	1.30×10^{-5}	0.1576	0.0969
$8 \times 24 \times 8$	3280	7.12×10^{-4}	5.50×10^{-7}	0.0410	0.0263
$8 \times 32 \times 8$	4368	6.95×10^{-4}	2.01×10^{-7}	0.0501	0.0334
$8 \times 36 \times 8$	4912	6.96×10^{-4}	1.87×10^{-7}	0.0504	0.0337
$10 \times 24 \times 10$	5540	3.81×10^{-4}	5.86×10^{-7}	0.0373	0.0291
$11 \times 22 \times 11$	6314	3.50×10^{-4}	7.20×10^{-7}	0.0246	0.0234
$12 \times 24 \times 12$	8376	2.43×10^{-4}	2.65×10^{-7}	0.0307	0.0309

4 Vacuum-field convergence

True convergence requires a fresh Hodge assembly and solve at every (n_r, n_θ, n_ζ) ; refining only the evaluation grid cannot change the discrete field. With dense verification, aligned L^2 decreases to 3.50×10^{-4} in the common-tracer $11 \times 22 \times 11$ record and to 2.43×10^{-4} in the $12 \times 24 \times 12$ field solve.

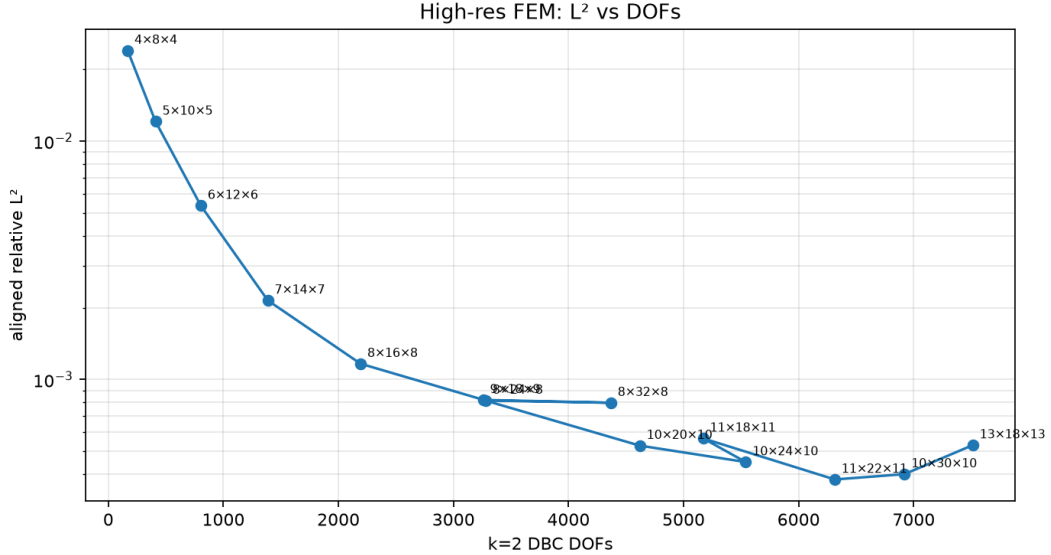


Figure 1: Aligned relative L^2 versus $k = 2$ Dirichlet DOFs for the dense-verified ladder (field convergence only). This ranking does *not* predict island fidelity: $8 \times 32 \times 8$ has worse L^2 than $12 \times 24 \times 12$ yet matches the SIMSOPT island (Section 5).

The isotropic ladder is consistent with a high-order spatial law $L^2 \sim h^p$, with fitted $p \approx 4.5$ for the $p = 3$ spline FEEC spaces. Island width instead tracks poloidal elements per lobe (Section 5): $a_{6,3}$ is most efficiently reduced by increasing n_θ , not by maximizing total DOFs.

5 Island-width convergence

5.1 Resonance location

All resolved grids place $\iota = 1/2$ near $\rho^* = 0.69\text{--}0.70$ and converge to a common local $\iota(\rho)$ profile. The $4 \times 8 \times 4$ mesh is the sole location outlier. Boundary resonant flux ($\sim 6 \times 10^{-9} B_{\text{rms}}$ on the GVEC $\rho = 1$ surface) and seed-density bias of the frozen $16\rho \times 6\theta$ protocol ($\sim 4\%$ on the $11 \times 22 \times 11$ detrended width) are both ruled out as drivers of residual width gaps on under-resolved grids.

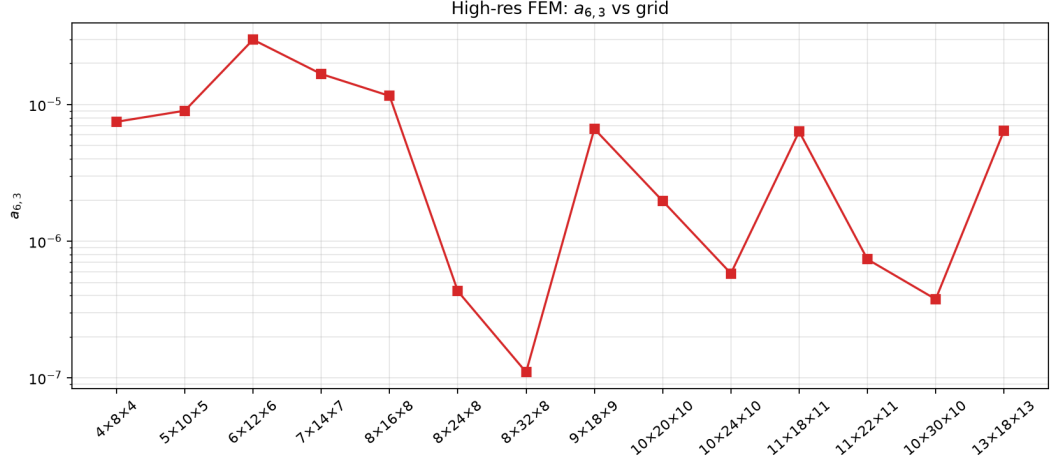
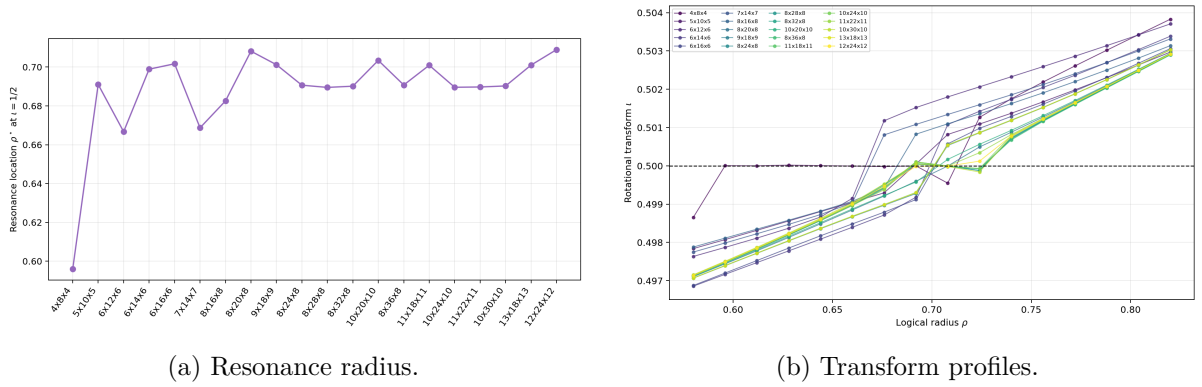


Figure 2: Resonant normal-error amplitude $a_{6,3}$. Poloidal refinement resolves the $m = 6$ error more efficiently than radial/toroidal enrichment at fixed n_θ .



(a) Resonance radius.

(b) Transform profiles.

Figure 3: Fixed-tracer convergence of resonance location and $\iota(\rho)$ through $\iota = 1/2$.

5.2 Resonant-pitch attribution and the relative width law

Across the saved fields, the complex resonant radial-pitch harmonic $\widehat{B^\rho/B^\zeta}_{6,1}$ at each grid's own $\iota = 1/2$ surface predicts the traced detrended width through the pendulum law

$$W = 4\sqrt{|c|/(2\pi |d\iota/d\rho|)},$$

where c is the complex Fourier coefficient of B^ρ/B^ζ (Figure 4). With phase $\psi = 2\pi(6\theta - \zeta)$ on the field-period coordinate and ι reported per full turn, the factor 2π is the correct shear normalization.

The shear cancels in the relative form

$$\frac{w_{\text{MRX}}}{w_{\text{SIMSOPT}}} = \sqrt{\frac{A_{\text{MRX}}}{A_{\text{SIMSOPT}}}},$$

which is the headline diagnostic (Figure 5). Trusted grids with at least ≈ 3.3 poloidal elements per lobe satisfy this identity to within a few percent; at $8 \times 32 \times 8$ (5.33 elements/lobe) both the amplitude ratio and the width ratio are unity against the corrected SIMSOPT reference $w = 0.0501$.

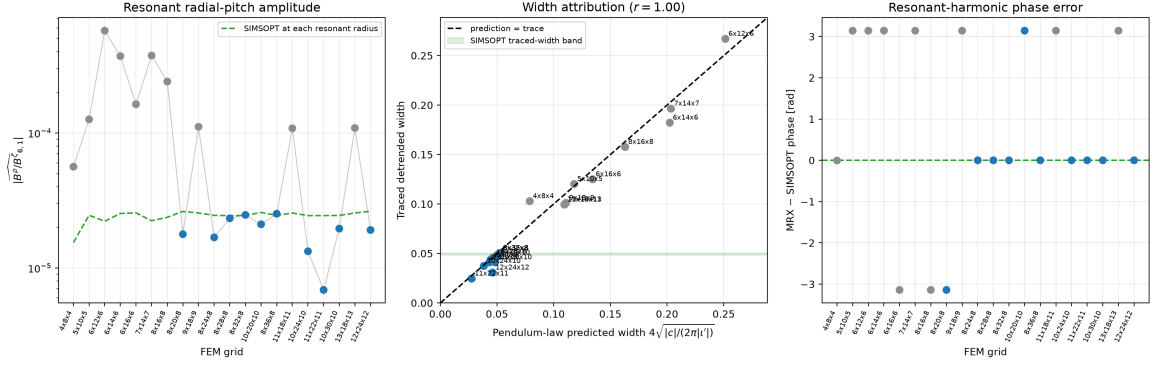


Figure 4: Width attribution. Resonant pitch amplitudes track poloidal resolution; predicted and traced detrended widths agree closely.

Aligned L^2 does *not* rank island fidelity. $8 \times 32 \times 8$ has 4368 DOFs and aligned $L^2 = 6.95 \times 10^{-4}$, worse than $12 \times 24 \times 12$ (8376 DOFs, $L^2 = 2.43 \times 10^{-4}$), yet $8 \times 32 \times 8$ matches the SIMSOPT island while the denser grid's truncated trace does not. DOF-ordered plots therefore look non-monotone even when the elements-per-lobe law is clean. Two grids are excluded from the headline fit: $13 \times 18 \times 13$ (dense/iterative M2 overlap 0.73, quarantined) and $12 \times 24 \times 12$ (truncated 1000-turn re-trace).

5.3 Fixed-aperture poloidal ladder

The decisive test is a six-point ladder at fixed aperture $n_\rho = n_\zeta = 8$, varying only n_θ (Table 2, Figure 5). Under-resolved $8 \times 16 \times 8$ (2.67 elements/lobe) produces a spurious numerical over-island. From 3.33 elements/lobe upward the amplitude and width ratios rise toward unity; at 5.33 and 6.00 elements/lobe both ratios are 1.00–1.03. This plateau confirms convergence rather than a crossing coincidence.

Dense verification is essential on this ladder: at $8 \times 36 \times 8$ the iterative null vector has M2 overlap only 0.185 with the dense eigenvector, yet the saved dense harmonic still yields the physical island. The iterative impurity is therefore a solver-health warning, not a disqualification of the dense-verified field.

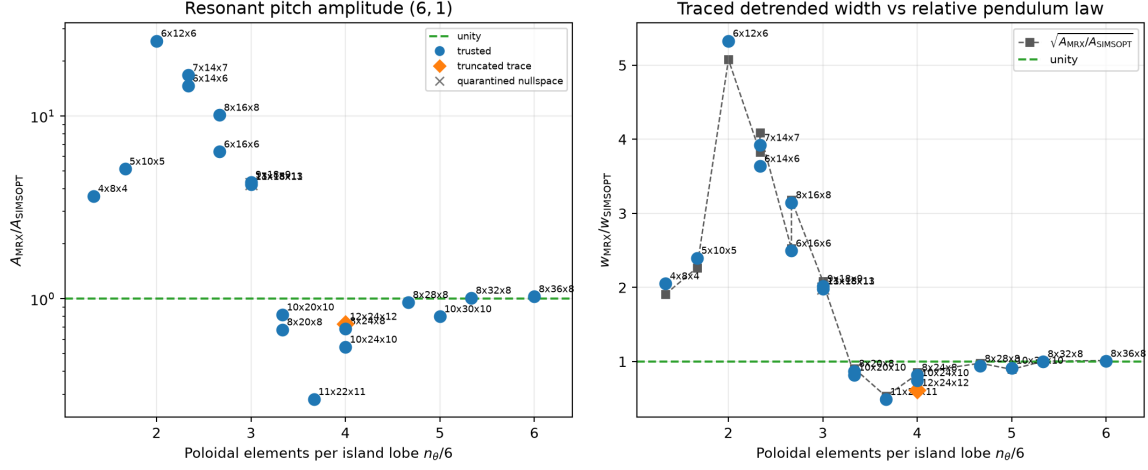


Figure 5: Island fidelity versus poloidal elements per lobe $n_\theta/6$. Left: resonant pitch amplitude ratio. Right: traced width ratio overlaid with $\sqrt{A_{MRX}/A_{SIMSOPT}}$. Under-resolved grids overshoot; the ratios approach unity near 5.3 elements/lobe. Orange markers are truncated traces; grey crosses are quarantined nullspaces.

Table 2: Fixed-aperture poloidal ladder ($n_\rho = n_\zeta = 8$, frozen $16\rho \times 6\theta \times 2000$ tracer). Width ratios use the corrected SIMSOPT reference $w = 0.0501$.

Grid	el/lobe	DOFs	$A_{MRX}/A_{SIMSOPT}$	w_{det}	$w/w_{SIMSOPT}$
$8 \times 16 \times 8$	2.67	2192	10.122	0.1576	3.146
$8 \times 20 \times 8$	3.33	2736	0.678	0.0437	0.872
$8 \times 24 \times 8$	4.00	3280	0.687	0.0410	0.818
$8 \times 28 \times 8$	4.67	3824	0.955	0.0470	0.938
$8 \times 32 \times 8$	5.33	4368	1.010	0.0501	1.001
$8 \times 36 \times 8$	6.00	4912	1.028	0.0504	1.006

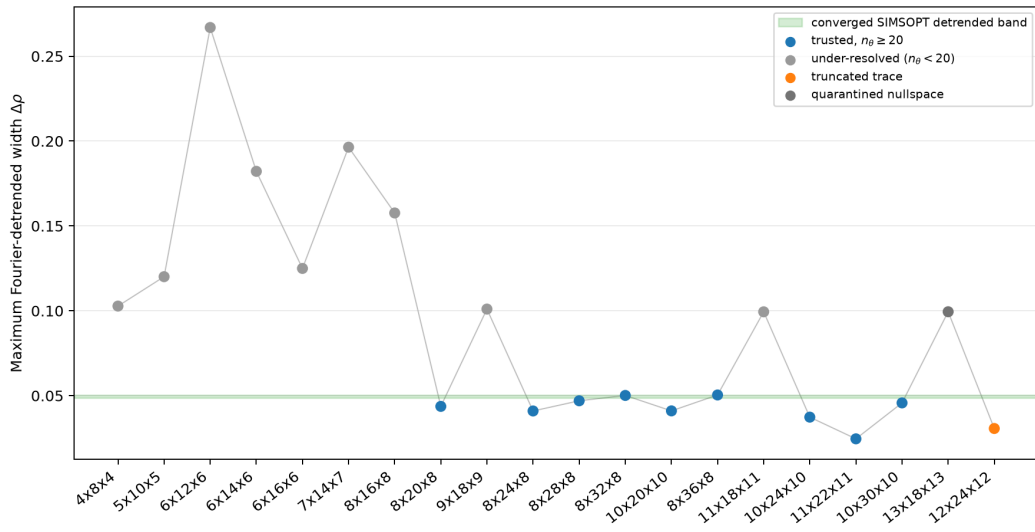


Figure 6: MRX maximum Fourier-detrended width compared with the converged SIMSOPT band. Poloidal resolution removes large numerical over-islands; $8 \times 32 \times 8$ and $8 \times 36 \times 8$ sit on the physical reference.

5.4 Manuscript Poincaré comparison

Figure 7 places the converged physical SIMSOPT chain beside the two highest points of the fixed-aperture ladder. Logical $\zeta = 0$ is not the cylindrical plane $\phi = 0$ used by SIMSOPT ($|\phi|$ reaches 3.4×10^{-2} rad on the wavy logical section); the physical row therefore embeds both fields through the same Newton-corrected $\phi = 0$ slice. SIMSOPT used 32 radial $\times$ 6 phase seeds; MRX used the frozen 16×6 protocol. The physical panels are a common coordinate embedding, not a claim that the two tracers recorded identical section events.

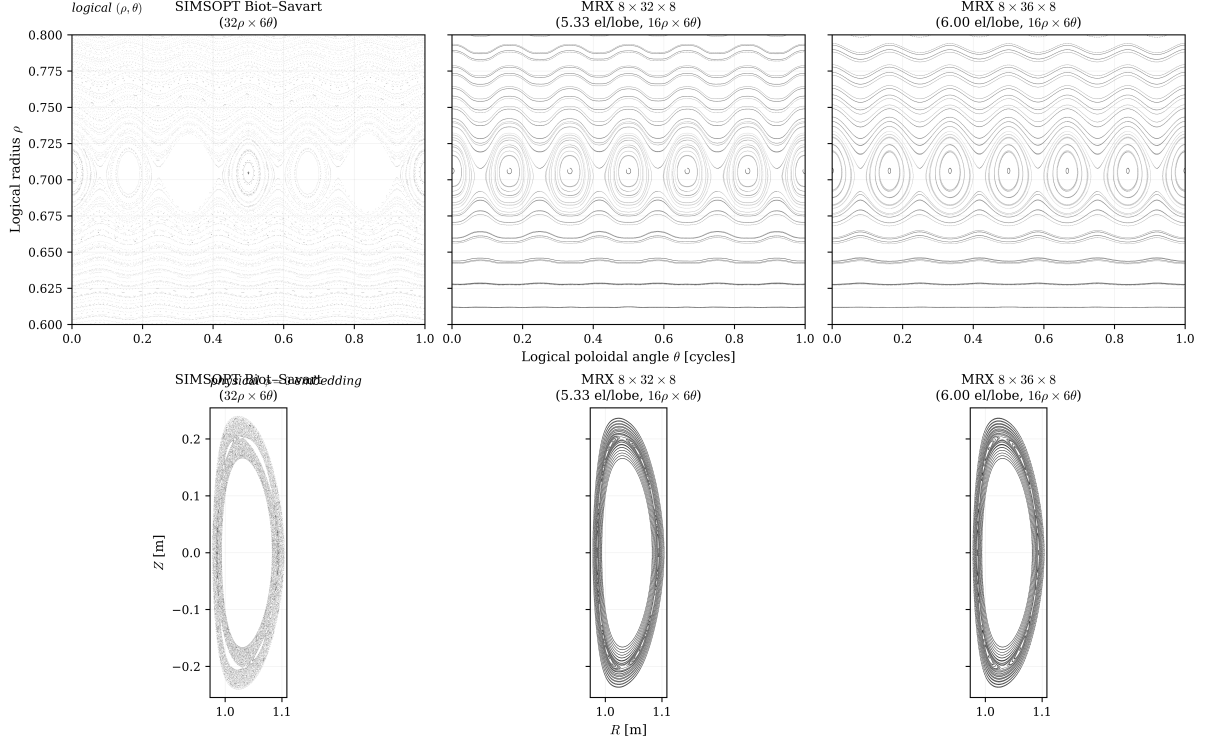


Figure 7: High-resolution Poincaré comparison. Top: logical (ρ, θ) island zoom. Bottom: trapped and passing orbits in a common physical $\phi = 0$ embedding. Columns are SIMSOPT Biot-Savart ($32\rho \times 6\theta$), MRX $8 \times 32 \times 8$ (5.33 elements/lobe), and MRX $8 \times 36 \times 8$ (6.00 elements/lobe). At these poloidal resolutions the MRX island chain matches the physical SIMSOPT width and location.

6 Code changes required

The following changes were required to make the L^2 and island-width demonstrations reliable. Each item names the function and the failure mode it fixed.

Dense `eigh` verification. Inexact shift-and-invert inverse iteration can return a vector with high scalar M_2 overlap but an impure non-harmonic component. Dense `eigh` on the assembled Hodge Laplacian replaces that vector with the exact discrete harmonic form before any diagnostic is computed. Without this step, field L^2 and island width both fail to converge monotonically.

Exact section crossings (`_exact_section_times`). Reparameterizing the field-line ODE so that $d\zeta/dt = 1$ makes Poincaré crossings land at known times. Adaptive-step integrators that detect zero-crossings of $\zeta - \zeta_{\text{section}}$ accumulate phase error over thousands of turns; the exact-time schedule removes that drift.

Bounded-chunk DiffraX batching (`_poincare_chunk_size`). Batched DiffraX Tsit5 is fast when every seed is healthy, but a single pathological seed forces tiny adaptive steps for the entire batch. Default chunking uses $\min(8, n_{\text{seeds}})$; the environment variable `MRX_POINCARE_CHUNK=1` forces per-seed integration for known bad grids (the truncated $12 \times 24 \times 12$ re-trace).

$\phi = 0$ slice Newton correction (`_physical_phi_zero_slice`, `_logical_to_phi_zero_rz`). SIMSOPT records the cylindrical plane $\phi = 0$; MRX was originally sampled at logical $\zeta = 0$. Inverting SIMSOPT hits against the wavy logical section imposed a spurious radial trend on the SIMSOPT O-points. Solving for the logical slice whose mapped points satisfy $\phi = 0$ removes that artifact and enables a fair physical overlay.

Fourier-detrended width (`_island_width_profile`). Raw $\rho_{q95} - \rho_{q05}$ conflates smooth surface-label shape with island width. Subtracting poloidal modes 1–4 from $\rho(\theta)$ before measuring the spread isolates the resonant deformation and yields the stable SIMSOPT reference $w_{\text{phys}} \approx 0.050$.

Field-based pitch shear. The helpers are `_iota_profile_from_physical_pitch` and `_local_shear_from_`. A five-point fit to a traced $\iota(\rho)$ profile is noisy near the resonance. Flux-surface-averaged logical pitch on a fine ρ grid gives a stable $d\iota/d\rho$ that is shared between SIMSOPT and MRX when forming the relative pendulum law.

Unified pendulum width (`pendulum_island_width`). Two earlier scripts used inconsistent shear normalizations (2π versus the poloidal mode number $m = 6$). The unified prefactor $W = 4\sqrt{|c|/(2\pi|d\iota/d\rho|)}$ matches the phase convention $\psi = 2\pi(6\theta - \zeta)$ with ι per full turn; the accidental $m = 6$ form is correct only to about 2.3%.

Circulation on the native single-period map. Evaluating $\oint B \cdot d\ell$ after artificially unfolding to the full torus introduced a gauge mismatch between SIMSOPT and MRX. Both circulations now use the native one-field-period map and the same interior loop $\rho = 0.6$, after which α and s^* agree.

Full-torus crossing cadence (`_physical_rotational_transform_profile`). Physical ι must advance by one full toroidal turn between returns, not one field period. Using the field-period cadence under-counted the transform by n_{fp} and misplaced the resonance.

M2-overlap reliability annotation. The aggregator marks grids with dense/iterative M2 overlap $\lesssim 0.75$ as quarantined nullspaces and truncated traces as outliers, so the headline elements-per-lobe fit is restricted to trusted records. Low overlap of the *iterative* vector alone is recorded as a solver-health caveat when the dense-verified field remains usable ($8 \times 36 \times 8$).

7 Conclusions

1. **Vacuum-field convergence is demonstrated.** Dense-verified aligned L^2 reaches the 10^{-4} range and follows high-order spatial convergence.
2. **The physical reference is $w_{\text{phys}} \approx 0.050$.** It is the seed-robust Fourier-detrended SIMSOPT width (0.0486–0.0503); no universal trapped-width reference is used.
3. **Island convergence is controlled by poloidal elements per lobe.** The relative law $w_{\text{MRX}}/w_{\text{SIMSOPT}} = \sqrt{A_{\text{MRX}}/A_{\text{SIMSOPT}}}$ holds for trusted grids. On the fixed-aperture ladder

both ratios reach unity at $8 \times 32 \times 8$ and remain there at $8 \times 36 \times 8$ (Table 2). Aligned L^2 does not rank island fidelity.

4. **Retain dense verification where topology matters.** The exact discrete harmonic form avoids the iterative impurity described in Section 2; quarantine grids with M2 overlap $\lesssim 0.75$ and caveat truncated traces.
5. **Resolve $m = 6$ poloidally before adding other DOFs.** Fixed-aperture enrichment in n_θ is the appropriate island resolution study; increasing radial and toroidal DOFs at fixed poloidal resolution does not converge the island.